q3

Submitting your work

At the end of class, submit the `.py` file you use for solving these problems to LEA.

When finished (or when time runs out) I will come and look at your progress and assess your work depending on how much you were able to finish.

Overview

You will analyze two datasets:

- the population of the United Kingdom and its constituent nations (`population.csv`)
- the co2 output of the United Kingdom and its constituent nations (`co2.csv`)

The data covered the years 2005-2014.

Your task is to write a Python program that imports the data, fits polynomial regression models, and predicts future CO2 emissions.

You are given the following dataset:

The data for both `.csv` files has the following form:

```
- - Year
  - 2005
  - 2006
  - etc.
- England
  - data for england in 2005
  - data for england in 2006
- England
  - data for Northern Ireland in 2005
  - data for Northern Ireland in 2006
  - etc.
- Scotland
  - data for Scotland in 2005
  - data for Scotland in 2006
  - etc.
- Wales
```

```
    - data for Wales in 2005
    - data for Wales in 2006
    - etc.
  - United Kingdom
    - data for UK in 2005
    - data for UK in 2006,
    - etc.
```

Your program must complete the following tasks:

1. (40%) Store the data

You will need to extract the data from each CSV, and create `numpy` arrays to store it.

That is, you should have separate variables storing numpy arrays for:

- all of the Years
- the population of Wales for each year
- the CO2 output in Wales for each year
- (etc., for each of the other nations)

2. (40%) Visualize the CO2 data

Create a scatter plot using `matplotlib` that displays the following:

- Labels for the X-axis (“Year”) and Y-axis (“CO2”)
- A scatter for the CO2 of each nation
  - (that is, the C02 output for each nation should be `Y`)
- A descriptive title
- A legend showing which points belong to which nation

FOCUS ON FINISHING ONE NATION AT A TIME

3. (10%) Create “per-capita” data and visualizations

Very similar to part 2, except: compute the “per-capita” values for each data point (NOTE: per-capita means per-person, so we want to know the CO2 output per person).

Once you have per-capita data, then make a scatter plot:

- Labels for the X-axis (“Year”) and Y-axis (“per capita CO2”)
- A scatter for the per-capita CO2 of each nation
  - (that is, the per-capita C02 output for each nation should be `Y`)

- A descriptive title
- A legend showing which points belong to which nation

4. (10%) Challenge

Use `numpy.polyfit()` and `numpy.polyval()` to generate predictions for the following:

- 2016-2025 CO2 data
- 2016-2025 population data

The choice of the degree of the polynomial is up to you! Pick which one makes the most sense to you.

Then go to https://ourworldindata.org/co2/country/united-kingdom. How close are your predictions? Write a comment in your python file with some discussion.